

String matching using FFT:

Suppose: text $\Rightarrow$ ABACA

Pattern $\Rightarrow$ ACA

for 'A' $\Rightarrow T = 10101$, $P = 101$

for alignment 0, we want $1*1 + 0*0 + 1*1 = 2$

that means $= T[0]P[0] + T[1]P[1] + T[2]P[2]$

for alignment 1, we want

$$= T[1]P[0] + T[2]P[1] + T[3]P[2]$$

$\vdots$

for general alignment s ,

we want
$$\sum_{j=0}^{m-1} T[s+j]P[j]$$

but direct convolution of two polynomial gives us

$$c[u] = \sum_i A[i]B[u-i]$$

$\Rightarrow$ We first reverse P , and let the final array be R .

then, we want
$$= \sum_{j=0}^{m-1} T[s+j]P[j]$$

here, $P[j] = R[m-1-j]$

convolution gives us,
$$c[u] = \sum_i T(i)R(m-1-(u-i))$$

Suppose, $m-1-u = -s$

$$\therefore u = s + m - 1$$

$$\therefore c[s+m-1] = \sum_i T(i) R(i-s)$$

$$\text{let, } j = i-s$$

$$\therefore c[s+m-1] = \sum_i T(s+j) R(j)$$

So, answer for the alignment starting from s^{th} char of T , is at $c[s+m-1]$.